

Problem Solving

1. Let X be observed number of successes in 20 Bernoulli trials. Consider $H_0 : \theta = 0.90$ vs. $H_1 : \theta = 0.60$. We accept H_0 if $X > 14$. What are type I and II errors?

2. Let $X_1, \dots, X_{10}$ be a random sample of size $n = 10$ from a Poisson distribution with mean θ . A critical region for testing $H_0 : \theta = 0.1$ against $H_1 : > 0.1$ is given by $Y = \sum_{i=1}^{10} X_i \geq 3$. What is the significance level of the test? If the critical region is defined by $\sum_{i=1}^{10} X_i \geq 4$, what is the new significance level?

3. Let $X_1, \dots, X_{n_1}$ be a random sample from $N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2}$ be a random sample from $N(\mu_2, \sigma^2)$. Consider $H_0 : \mu_1 = \mu_2$. Discuss the test against $H_1 : \mu_1 > \mu_2$ (or $H_1 : \mu_1 < \mu_2$, or $H_1 : \mu_1 \neq \mu_2$).

